Assignment 1: Math1014

September 8, 2026

1 Linear Algebra

1.1 Question 1

we are given the vector space

$$V_N = \operatorname{span}\{\cos(2\pi kt), \sin(2\pi kt) \mid 0 \leq k \leq N\}.$$

(a)

two functions are linearly independent if the only scalars a, b satisfying

$$a\cos(2\pi t) + b\sin(2\pi t) = 0 \quad \text{for all } t \in [0, 1]$$

are $a = b = 0$. if we can find even one nontrivial choice of t that forces $a = 0$, and another that forces $b = 0$, we're done.
- at $t = 0$,

$$a\cos(0) + b\sin(0) = a(1) + b(0) = a = 0$$

so a must be 0
- at $t = \frac{1}{4}$

$$a\cos\left(\frac{\pi}{2}\right) + b\sin\left(\frac{\pi}{2}\right) = a(0) + b(1) = b = 0$$

so b must be 0 too
since the equation forces $a = b = 0$, the only linear combination equal to the zero function is the trivial one. hence $\cos(2\pi t)$ and $\sin(2\pi t)$ are **linearly independent** in V_1.

(b)

we need the dimension of V_1
the definition says

$$V_N = \operatorname{span}\{\cos(2\pi kt), \sin(2\pi kt) \mid 0 \leq k \leq N\}$$

if $N = 1$, then k can be either 0 or 1.
so we get

$$V_1 = \text{span}\{\cos(0), \sin(0), \cos(2\pi t), \sin(2\pi t)\}$$

now, $\cos(0) = 1$ and $\sin(0) = 0$
so the span becomes:

$$V_1 = \text{span}\{1, \cos(2\pi t), \sin(2\pi t)\}$$

the zero function doesn't add anything to the span, so we can ignore it
we therefore have three functions:

$$1, \cos(2\pi t), \sin(2\pi t)$$

these are linearly independent, so they form a basis for V_1.
therefore, $\boxed{dim(V_1) = 3}$

(c)

finding the dimension of V_2,
when $N = 2$, the possible values of k are $0, 1, 2$
therefore,

$$V_2 = \text{span}\{\cos(0), \sin(0), cos(2\pi t), sin(2\pi t), cos(4\pi t), sin(4\pi t)\}$$

again, $\cos(0) = 1$ and $\sin(0) = 0$, so (leaving out the 0 terms),

$$V_2 = \text{span}\{1, cos(2\pi t), sin(2\pi t), cos(4\pi t), sin(4\pi t)\}$$

there are now 5 independent functions. therefore,

$$\boxed{dim(V_2) = 5}$$

for a general N, we always have:

- one constant function 1,
- N cosine functions corresponding to $k = 1, ..., N$,
- n sine functions coreresponding to $K = 1, ..., N$

so, the total number of basis functions is

$$1 + N + N$$

therefore,

$$\boxed{dim(V_N) = 2N + 1}$$

1.2 Question 2

we have $S : V_N \to R_m$ defined by

$$S(f) = \begin{bmatrix} f(t_1) \\ f(t_2) \\ \vdots \\ f(t_m) \end{bmatrix}$$

(a)

we are given $m = 3$ with sampling points $t_1 = 0$, $t_2 = 0.5$, $t_3 = 1$ and

$$f(t) = 3 + \cos(2\pi t)$$

since S records the value of f at each sampling point,

$$S(f) = \begin{bmatrix} f(0) \\ f(0.5) \\ f(1) \end{bmatrix}$$

- at $t = 0$,

  $$f(0) = 3 + \cos(2\pi(0))$$
  $$f(0) = 3 + \cos(0)$$

  since $\cos(0) = 1$,

  $$f(0) = 4$$

- at $t = 0.5$,

  $$f(0.5) = 3 + \cos(2\pi(0.5))$$
  $$f(0.5) = 3 + \cos(\pi)$$

  since $\cos(\pi) = -1$,

  $$f(0.5) = 3 - 1 = 2$$

- at $t = 1$,

  $$f(1) = 3 + \cos(2\pi(1))$$
  $$f(1) = 3 + \cos(2\pi)$$

  since $\cos(2\pi) = 1$,

  $$f(1) = 3 = 1 = 4$$

putting all these together into the matrix, we get

$$\boxed{S(f) = \begin{bmatrix} 4 \\ 2 \\ 4 \end{bmatrix}}$$

(b)

now we have

$$g(t) = \sin(12\pi t)$$

again,

$$S(g) = \begin{bmatrix} g(0) \\ g(0.5) \\ g(1) \end{bmatrix}$$

- at $t = 0$,
$$g(0) = \sin(12\pi(0)) = \sin(0) = 0$$
- at $t = 05$,
$$g(0.5) = \sin(12\pi(0.5)) = \sin(6\pi) = 0$$
- at $t = 1$,
$$g(1) = \sin(12\pi(1)) = \sin(12\pi) = 0$$

putting all these together into the matrix, we get

$$\boxed{S(f) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}$$

(c)

for a function to be a linear map, we need

$S(af + bg) = aS(f) + bS(g)$ for any functions $f, g \in V_N$ and scalars $a, b \in \mathbb{R}$

starting with the LHS,

$$S(f) = \begin{bmatrix} (af + bg)(t_1) \\ (af + bg)(t_2) \\ \vdots \\ (af + bg)(t_m) \end{bmatrix}$$

we know that at each sample point t_i, the addition and scalar multiplication is

$$(af + bg)(t_i) = af(t_i) + bg(t_i)$$

therefore,

$$S(f) = \begin{bmatrix} af(t_1) + bg(t_1) \\ af(t_2) + bg(t_2) \\ \vdots \\ af(t_m) + bg(t_m) \end{bmatrix}$$

separating into two vectors:

$$S(f) = a \begin{bmatrix} f(t_1) \\ f(t_2) \\ \vdots \\ f(t_m) \end{bmatrix} + b \begin{bmatrix} g(t_1) \\ g(t_2) \\ \vdots \\ g(t_m) \end{bmatrix}$$

we can see that these two vectors are exactly $S(f)$ and $S(g)$
therefore,

$$S(af + bg) = aS(f) + bS(g)$$

hence,

$$\boxed{\text{S is linear}}$$

(d)

an element of V_N can be represented as a vector in $\mathbb{R}^{2N+1}$
from question 1, $dim(V_N) = 2N + 1$, so $[f]_{\mathcal{B}} \in \mathbb{R}^{2N+1}$.
so there are $2N+1$ functions in this basis, meaning $[f]_{\mathcal{B}}$ has $2N+1$ components
S takes f and evaluates it at m sample points

$$S(f) = \begin{bmatrix} f(t_1) \\ f(t_2) \\ \vdots \\ f(t_m) \end{bmatrix}$$

therefore, $S(f) \in \mathbb{R}^m$. so the output has m components
so we need a matrix A such that

$$\text{matrix } A \times \text{input vector} = \text{output vector}$$

therefore, A needs to take a vector with $2N+1$ entries and turn it into a vector with m entries.
this means A needs $2N + 1$ columns (one for each basis function) and m rows (one for each sample point).
therefore,

$$\boxed{A \in \mathbb{R}^{m \times (2N+1)}}$$

(e)

we know from 2(d) that

$$A \in \mathbb{R}^{m \times (2N+1)}$$

so A has m rows and $2N + 1$ columns.
the rank of a matrix cannot be bigger than the number of rows or the number of columns.

therefore,

$$rank(A) \leq m$$

and

$$rank(A) \leq 2N + 1$$

so the upper bound is the smaller of these two numbers:

$$\boxed{rank(A) \leq \min(m, 2N + 1)}$$

this is because the rank tells us how many independent rows or columns there are, so we cannot have more independent rows than m, or more independent columns than $2N + 1$.

(f)

we can use the rank-nullity theorem:

$$rank(A) + nullity(A) = \text{number of columns of } A$$

from 2(d), A has $2N + 1$ columns, so

$$rank(A) + nullity(A) = 2N + 1$$

rearranging,

$$nullity(A) = 2N + 1 - rank(A)$$

from part (e),

$$rank(A) \leq \min(m, 2N + 1)$$

so the smallest possible nullity occurs when the rank is as large as possible. therefore,

$$nullity(A) \geq 2N + 1 - \min(m, 2N + 1)$$

so,

$$\boxed{nullity(A) \geq \max(0, 2N + 1 - m)}$$

if $m < 2N + 1$, then

$$\boxed{nullity(A) \geq 2N + 1 - m > 0}$$

(g)

from part(f) we know that

$$nullity(A) \geq 2N + 1 - m.$$

we want

$$dim(null(A)) > 0.$$

this will definitely happen when

$$2N + 1 - m > 0.$$

so,

$$2N + 1 > m.$$

therefore,

$$\boxed{m < 2N + 1}$$

leads to

$$\boxed{dim(null(A)) > 0}.$$

in other words, if we have fewer sample points than the number of basis functions, there must be some non-zero functions that get mapped to the zero vector.

(h)

2 Calculus

2.1 Question 3

a

b

c

2.2 Question 4

a

b

c